

System of Linear Equations

A collection of one or more linear equations with same number of unknowns $x_1, \dots, x_n$

e.g. $2x_1 - x_2 + 1.5x_3 = 8$ ①
 $x_1 - 4x_3 = -7$ ②

solutions: $(x_1 = -7, x_2 = -22, x_3 = 0)$, $(x_1 = 1, x_2 = -3, x_3 = 2)$,
solution set:

$$\begin{aligned} \textcircled{1} - 2 \times \textcircled{2} &\Rightarrow -x_2 + 1.5x_3 - (-8x_3) = 8 - (-14) \\ &\Rightarrow -x_2 + 9.5x_3 = 22 \\ &\Rightarrow x_2 = 9.5x_3 - 22 \end{aligned}$$

$$\textcircled{2} \Rightarrow x_1 = 4x_3 - 7$$

Then the solution set is the set of (x_1, x_2, x_3) such that
 $x_1 = 4x_3 - 7$ and $x_2 = 9.5x_3 - 22 \Rightarrow$ infinitely many solutions

A system of equations can have

{	1. no solution	inconsistent
	2. exactly one solution	consistent
	3. infinitely many solution	